Weekly Progress Report

Name: Devwrat Gupta
Domain: Python Programming
Date of submission: 29/06/2026

Week Ending: 03

I. Overview:

This week, I learned the basics of NumPy and Pandas libraries in Python. I explored how NumPy is used for working with arrays and mathematical operations, while Pandas is useful for handling tabular data. I also continued revising my File Organizer project to connect these concepts with practical programming.

II. Achievements:

1. USC_TIA Familiarization:
   - Completed the learning modules related to NumPy and Pandas.
   - Explored the examples provided during the internship sessions to understand their basic usage.

2. Python Project Contributions:

Name of the project:- File Organizer

- Reviewed the File Organizer project developed in the previous weeks.
- Continued practicing Python programming by understanding the project structure and logic.
- Focused on strengthening Python fundamentals instead of adding new project features.
- Improved the project by adding support for multiple file extensions and handling duplicate files.

3.Learning Python:

- Learned the basics of the NumPy library and its common operations.
- Practiced concepts such as array reshaping, indexing, slicing, replacing elements, and basic mathematical functions.
- Learned the fundamentals of the Pandas library, including Series, DataFrame, and importing/exporting data.

III. Challenges:

1. Understanding NumPy :

- Initially found NumPy arrays different from normal Python lists.
- Took some time to understand reshaping arrays and indexing in multiple dimensions.

2. Understanding Pandas :

- Faced some difficulty in understanding the difference between Series and DataFrame.
- Practiced examples to become more familiar with handling tabular data.

IV. Learning Resources:

1. USC_TIA Documentation:

- Referred to the internship videos and learning materials provided for Week 3.
- Studied the examples related to NumPy and Pandas operations.

2. Python Learning Resources:

- Practiced sample programs using NumPy arrays and Pandas DataFrames.
- Explored basic examples to understand data manipulation and analysis.

V. Next Week's Goals:

1. Python Learning:

- Learn upcoming Python concepts and apply them in practical examples.
- Improve coding skills through regular practice.

2.Python Project Development:

- Continue reviewing and improving the File Organizer project.
- Explore ways to make the code more organized and efficient.
- Apply newly learned Python concepts wherever suitable in future projects.

VI. Additional Comments:

This week's learning introduced me to NumPy and Pandas, which are widely used in Python for data processing and analysis. Although these libraries are new to me, practicing simple examples helped me understand their basic purpose and increased my confidence in learning advanced Python topics.